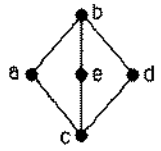
that $P(i)$ is maximized and following the C links.

SECTION 10.7 Planar Graphs

As with Euler and Hamilton circuits and paths, the topic of planar graphs is a classical one in graph theory. The theory (Euler's formula, Kuratowski's theorem, and their corollaries) is quite beautiful. It is easy to ask extremely difficult questions in this area, however—see Exercise 27, for example. In practice, there are very efficient algorithms for determining planarity that have nothing to do with Kuratowski's theorem, but they are quite complicated and beyond the scope of this book. For the exercises here, the best way to show that a graph is planar is to draw a planar embedding; the best way to show that a graph is nonplanar is to find a subgraph homeomorphic to K_5 or $K_{3,3}$. (Usually it will be $K_{3,3}$.)

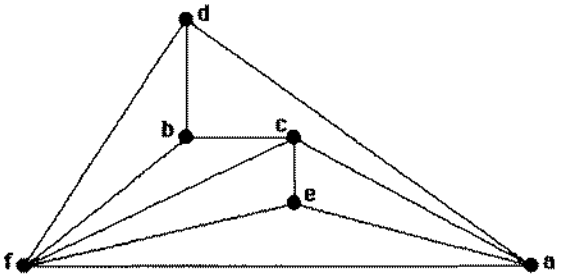
1. The question is whether $K_{5,2}$ is planar. It clearly is so, since we can draw it in the xy-plane by placing the five vertices in one part along the x-axis and the other two vertices on the positive and negative y-axis.

3. For convenience we label the vertices a, b, c, d, e, starting with the vertex in the lower left corner and proceeding clockwise around the outside of the figure as drawn in the exercise. This graph is just $K_{2,3}$; the picture below shows it redrawn by moving vertex c down.



5. This is $K_{3,3}$, with parts $\{a, d, f\}$ and $\{b, c, e\}$. Therefore it is not planar.

7. This graph can be untangled if we play with it long enough. The following picture gives a planar representation of it.



9. If one has access to software such as *The Geometer's Sketchpad*, then this problem can be solved by drawing the graph and moving the points around, trying to find a planar drawing. If we are unable to find one, then we look for a reason why—either a subgraph homeomorphic to K_5 or one homeomorphic to $K_{3,3}$ (always try the latter first). In this case we find that there is a homeomorphic copy of $K_{3,3}$, with vertices b, g, and i in one set and a, f, and h in the other; all the edges are there except for the edge bh, and it is represented by the path beh.

11. We give a proof by contradiction. Suppose that there is a planar representation of K_5, and let us call the vertices v_1, v_2, ..., v_5. There must be an edge from every vertex to every other. In particular, v_1, v_2, v_3, v_4, v_5, v_1 must form a pentagon. The pentagon separates the plane into two regions, an inside and an outside. The edge from v_1 to v_3 must be present, and without loss of generality let us assume it is drawn on the inside. Then there is no way for edges $\{v_2, v_4\}$ and $\{v_2, v_5\}$ to be in the inside, so they must be in the outside region. Now this prevents edges $\{v_1, v_4\}$ and $\{v_3, v_5\}$ from being on the outside. But they cannot both be on the inside without crossing. Therefore there is no planar representation of K_5.

13. We apply Euler's formula $r = e - v + 2$. Here we are told that $v = 6$. We are also told that each vertex has degree 4, so that the sum of the degrees is 24. Therefore by the handshaking theorem there are 12 edges, so $e = 12$. Solving, we find $r = 8$.

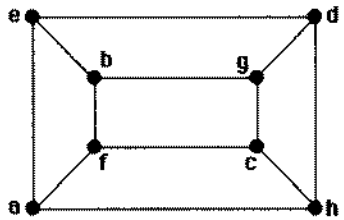
15. The proof is very similar to the proof of Corollary 1. First note that the degree of each region is at least 4. The reason for this is that there are no loops or multiple edges (which would give regions of degree 1 or 2) and no simple circuits of length 3 (which would give regions of degree 3); and the degree of the unbounded region is at least 4 since we are assuming that $v \geq 3$. Therefore we have, arguing as in the proof of Corollary 1, that $2e \geq 4r$, or simply $r \leq e/2$. Plugging this into Euler's formula, we obtain $e - v + 2 \leq e/2$, which gives $e \leq 2v - 4$ after some trivial algebra.

17. The proof is exactly the same as in Exercise 15, except that this time the degree of each region must be at least 5. Thus we get $2e \geq 5r$, which after the same algebra as before, gives the desired inequality.

19. **a)** If we remove a vertex from K_5, then we get K_4, which is clearly planar.
b) If we remove a vertex from K_6, then we get K_5, which is not planar.
c) If we remove a vertex from $K_{3,3}$, then we get $K_{3,2}$, which is clearly planar.
d) We assume the question means "Is it the case that for every v, the removal of v makes the graph planar?" Then the answer is no, since we can remove a vertex in the part of size 4 to leave $K_{3,3}$, which is not planar.

21. This graph is planar and hence cannot be homeomorphic to $K_{3,3}$.

23. The instructions are really not fair. It is hopeless to try to use Kuratowski's theorem to prove that a graph *is* planar, since we would have to check hundreds of cases to argue that there is no subgraph homeomorphic to K_5 or $K_{3,3}$. Thus we will show that this graph is planar simply by giving a planar representation. Note that it is Q_3.



25. This graph is nonplanar, since it contains $K_{3,3}$ as a subgraph: the parts are $\{a, g, d\}$ and $\{b, c, e\}$. (Actually it contains $K_{3,4}$, and it even contains a subgraph homeomorphic to K_5.)

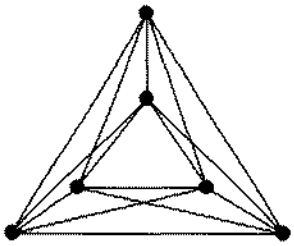
27. This is an extremely hard problem. We will present parts of the solution; the reader should consult a good graph theory book, such as Gary Chartrand, Linda Lesniak and Ping Zhang's *Graphs & Digraphs*, fifth edition (Chapman & Hall/CRC Press, 2011), for references and further details.

First we will state, without proof, what is known about crossing numbers for complete graphs (much is still not known about crossing numbers). If $n \leq 10$, then the crossing number of K_n is given by the following product

$$\frac{1}{4}\left\lfloor \frac{n}{2} \right\rfloor \left\lfloor \frac{n-1}{2} \right\rfloor \left\lfloor \frac{n-2}{2} \right\rfloor \left\lfloor \frac{n-3}{2} \right\rfloor.$$

Thus the answers for parts **(a)**, **(b)**, and **(c)** are 1, 3, and 9, respectively. The figure below shows K_6 drawn in the plane with three crossings, which at least proves that the crossing number of K_6 is at most 3. The

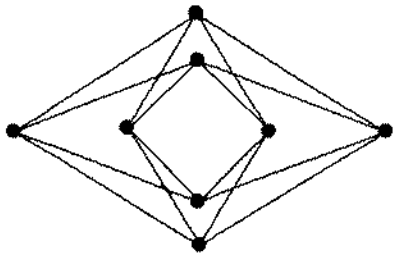
proof that it is not less than 3 is not easy. The embedding of K_5 with one crossing can be seen in this same picture, by ignoring the vertex at the top.



Second, for the complete bipartite graphs, what is known is that if the smaller of m and n is at most 6, then the crossing number of $K_{m,n}$ is given by the following product

$$\left\lfloor \frac{m}{2} \right\rfloor \left\lfloor \frac{m-1}{2} \right\rfloor \left\lfloor \frac{n}{2} \right\rfloor \left\lfloor \frac{n-1}{2} \right\rfloor.$$

Thus the answers for parts **(d)**, **(e)**, and **(f)** are 2, 4, and 16, respectively. The figure below shows $K_{4,4}$ drawn in the plane with four crossings, which at least proves that the crossing number of $K_{4,4}$ is at most 4. The proof that it is not less than 4 is, again, difficult. It is also easy to see from this picture that the crossing number of $K_{3,4}$ is at most 2 (by ignoring the top vertex).



29. Let us follow the hint, and draw all the edges with straight line segments. This clearly produces a drawing of $K_{m,n}$. We will show that the number of crossings is $mn(m-2)(n-2)/16$, and that will complete the proof. (Incidentally, it is not known whether this upper bound is actually the crossing number. No one has found an embedding with fewer crossings, but only in the case in which the smaller of m and n is at most 6 has it been proved that it cannot be done. See the comments in the solution to Exercise 27.) In order to count the crossings, it is enough to count the crossings occurring in the first quadrant and multiply by 4. Let us label the points on the positive x-axis with the numbers 1 through $m/2$, and those on the y-axis with the numbers 1 through $n/2$. If we choose any two distinct numbers, say a and b with $a < b$, from 1 to $m/2$, and any two distinct numbers, say r and s with $r < s$, from 1 through $n/2$, then we get exactly one crossing in our graph, namely between the edges as and br. (There is no crossing between ar and bs.) So the number of crossings in the first quadrant is the same as the number of ways to make these choices, which is clearly $C(m/2,2) \cdot C(n/2,2)$. So the total number of crossings is 4 times this quantity, namely

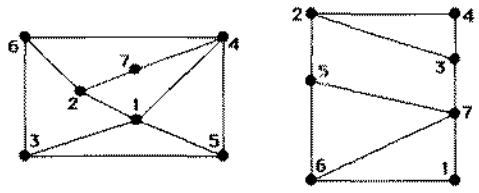
$$4 \cdot C(m/2,2) \cdot C(n/2,2) = 4 \cdot \frac{\frac{m}{2}\left(\frac{m}{2}-1\right)}{2} \cdot \frac{\frac{n}{2}\left(\frac{n}{2}-1\right)}{2},$$

which easily simplifies to

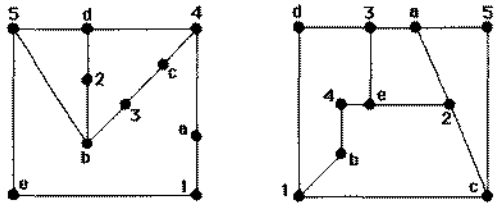
$$\frac{mn(m-2)(n-2)}{16}.$$

31. Each of these graphs is nonplanar; the first three contain K_5, and the last three contain $K_{3,3}$. Thus if we can show how to draw each of the graphs in two planes, then we will have shown that the thickness is 2 in each

case. The following picture shows that K_7 can be drawn in 2 planes, so this takes care of part **(a)**, part **(b)**, and part **(c)**.



The following picture shows that $K_{5,5}$ can be drawn in 2 planes, so this takes care of part **(d)**, part **(e)**, and part **(f)**.



33. The formula is certainly valid for $n \leq 4$, so let us assume that $n > 4$. By Exercise 32, the thickness of K_n is at least

$$\frac{C(n,2)}{3n-6} = \frac{n(n-1)/2}{3n-6} = \frac{n(n-1)}{6(n-2)} = \frac{1}{6}\left(n+1+\frac{2}{n-2}\right)$$

rounded up. Since this quantity is never an integer, it equals one more than itself rounded down, namely

$$\frac{1}{6}\left(n+1+\frac{2}{n-2}\right)+1 = \frac{n+7}{6}+\frac{2}{6(n-2)}$$

rounded down. The last term can be ignored: it is always less than $1/6$ and therefore will not influence the rounding process (since the first term has denominator 6). Thus we have proved that the thickness of K_n is at least $\lfloor (n+7)/6 \rfloor$.

35. This follows immediately from Exercise 34, since $K_{m,n}$ has mn edges and $m+n$ vertices and, being bipartite, has no triangles.

37. We can represent the surface of a torus with a rectangle, thinking of the right-hand edge as being equal to the left-hand edge, and the top edge as being equal to the bottom edge. For example, if we travel out of the rectangle across the right-hand edge about a third of the way from the top, then we immediately reenter the rectangle across the left-hand edge about a third of the way from the top. The picture below shows $K_{3,3}$ drawn on this surface. Note that the edges that seem to leave the rectangle really reenter it from the opposite side.

